# Simulation Study: Base Model and Sensitivity Scenarios

A Bayesian Framework for Causal Inference with Error-Prone Environmental Exposures

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## 1 Motivation

The simulation reflects the two-level structure of the application: area-level count health outcomes for participants nested within Forward Sortation Areas (FSA or the first three characters of a Canadian postal code), with area-level exposure assigned from a spatiotemporal prediction model.

## 2 Notation

- $i = 1, \ldots, I$: area index (FSAs)
- $A_i$: latent true area-level $PM_{2.5}$ exposure, unobserved
- $Z_i$: observed error-prone surrogate from a spatiotemporal prediction model (e.g. CanOSSEM)
- $\mathbf{X}_i = (X_{i1}, \ldots, X_{i4})^\top$: area-level confounders affecting both exposure and outcome
- $D_i$: area-level calibration covariate explaining prediction error magnitude in $Z_i$ (e.g. distance to nearest monitoring station, area deprivation)
- $Y_i$: area-level count health outcome (number of incident depression cases)
- $N_i$: area population size, used as Poisson offset
- $\phi_i$: spatial random effect on the latent exposure surface, following the CAR prior in equation (1)
- $\mathbf{W}$: $I \times I$ binary adjacency matrix ($W_{ij} = 1$ if areas $i$ and $j$ share a border)
- $\mathbf{D}_W = \text{diag}(\mathbf{W1})$: diagonal degree matrix (number of neighbours per area)
- $\rho_A \in [0, 1)$: spatial correlation in the exposure CAR prior
- $\sigma^2_\phi$: marginal variance of the spatial random effect $\phi_i$

As in the thesis proposal, $\mathbf{X}_i$ controls confounding while $D_i$ indexes the measurement error mechanism. $D_i$ may overlap with $\mathbf{X}_i$ but is conceptually distinct.

# 3 Key Assumptions

The simulation satisfies the following identification assumptions from the thesis proposal.

- **Non-differential surrogate:** $Y_i \perp Z_i \mid A_i, \mathbf{X}_i, D_i$. Once we condition on the true exposure and covariates, the surrogate carries no additional information about the outcome. Satisfied by construction since $Y_i$ is generated from $A_i$ only and $Z_i$ never appears in the outcome equation.
- **Transportability:** The error relationship is assumed transportable to the target population. When direct calibration data are unavailable, inference about $A_i$ is more model-dependent; this is examined in sensitivity scenario S2.

# 4 Base Model

## 4.1 Spatial Structure and CAR Prior

Areas are arranged on a $\sqrt{I} \times \sqrt{I}$ synthetic grid with rook contiguity (shared edges only). For this simulation the base value is $I = 400$ (a $20 \times 20$ grid); $I = 196$ and $I = 900$ are used as smaller and larger cases.

Following thesis proposal Section 4.3, spatial dependence enters only through the latent exposure model via $\phi_i$. The spatial random effect follows the Conditional Autoregressive (CAR) prior:

$$\boldsymbol{\phi} \mid \rho_A, \sigma_\phi^2 \sim \mathcal{N}\big(\mathbf{0},\ \sigma_\phi^2(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}\big)\,. \tag{1}$$

- $\phi_i$ is a spatially structured random effect: neighbouring areas have similar $\phi_i$ values, inducing correlation in their true exposures $A_i$
- The precision matrix $(\mathbf{D}_W - \rho_A \mathbf{W})$ is sparse with adjacent pairs having off-diagonal entry $-\rho_A$ and non-adjacent pairs having zero
- $\rho_A = 0$: areas are spatially independent; $\rho_A \to 1$: strong smoothing across neighbours
- Spatial correlation in the outcome is induced indirectly through $A_i$; no separate outcome random effect is added

## 4.2 Covariates and Population Sizes

- **Area confounders:** $X_{i1}, X_{i3}, X_{i4} \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$; $X_{i2} = (G-2)/\sqrt{2}$ with $G \sim \text{Gamma}(2,1)$ is right-skewed (skewness 1.41). All four have mean 0 and variance 1, so the exposure-variance bookkeeping below is unchanged. The skewed one is the confounder with the smallest outcome coefficient, which limits how much its heavy tail inflates the gap between a dataset's covariate-averaged ERF and the population value
- **Population sizes:** $\log(N_i) \sim \mathcal{N}(8.5, 1.2^2)$, giving median $\approx$ 4,900, 5th percentile $\approx$ 600, 95th percentile $\approx$ 40,000. Motivated by Statistics Canada 2021 Census: Ontario has $\approx$ 500 FSAs with total population 14,223,942, ranging from dense urban FSAs in Toronto to sparse rural FSAs in Northern Ontario.

### 4.3 Calibration Covariate $D_i$

$D_i$ captures area characteristics that determine prediction error in $Z_i$.

**Base model: scalar $D_i$.** We generate a raw covariate $D_i^* \overset{\text{iid}}{\sim} \text{Gamma}(2, 1)$ (mean 2, variance 2) representing distance to the nearest monitoring station, then standardise:

$$D_i = \frac{D_i^* - 2}{\sqrt{2}},$$

so that $D_i$ has mean 0 and variance 1. Standardisation prevents numerical blow-up in the exponential error variance. The interpretation of $\eta_1$ demonstrates that a one standard deviation increase in monitor distance multiplies the error variance by $\exp(\eta_1)$.

- Distance is strictly positive and right-skewed and hence a Gamma distribution naturally showcases this. Most Ontario FSAs are moderately close to one of the 34 monitoring stations, but some rural areas are very far.
- **Motivation:** Gryparis et al. (2009) and Szpiro et al. (2011) show that prediction error variance is primarily determined by distance to the monitor network. MELONS (Evangelopoulos et al., 2026) identified distance, deprivation, and housing type as empirical determinants of $PM_{2.5}$ measurement error in London, UK

**Extension: vector $D_i$.** In a full implementation $D_i$ would be a vector with components such as: $D_{i1} \sim \text{Gamma}(2, 1)$ (distance to monitor, standardised), $D_{i2} \sim \mathcal{N}(0, 1)$ (income and employment indicators as a standardised score), $D_{i3} \sim \text{Beta}(2, 2)$ (proportion detached housing). The error model would be $\omega^2(\mathbf{D}_i) = \exp(\eta_0 + \eta_1 D_{i1} + \eta_2 D_{i3})$. Left as a future extension.

### 4.4 Latent True Exposure

The latent true exposure is generated from:

$$A_i \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}(10 - 0.75X_{i1} + 0.75X_{i2} - 0.75X_{i3} + 0.75X_{i4} + \phi_i,\ 2.25)\,, \tag{2}$$

with $\sigma_A = 1.5$, where $\phi_i$ is the spatial random effect following the CAR prior defined in equation (1) with parameters $(\rho_A, \sigma_\phi^2)$. The spatial random effect $\phi_i$ induces correlation in $A_i$ across neighbouring areas: areas with $\rho_A$ close to 1 have exposures that vary smoothly across the grid, reflecting a regional wildfire plume, while $\rho_A = 0$ gives spatially independent exposures. The exposure is not truncated: with a mean of 10 and a total SD of about 2.4, the Normal model puts essentially no mass below zero (no negative $A_i$ in 60 test datasets).

- **Mean = 10 $\mu$g/m$^3$:** A cumulative wildfire-smoke exposure level, chosen high enough relative to the spread that $A_i$ and $Z_i$ are essentially never negative, so no truncation or censoring is needed (about 5 to 15 $\mu$g/m$^3$ covers the 1st to 99th percentile)
- $\sigma_A = 1.5$**:** Residual SD; with the covariate and spatial components the total SD of $A_i$ is about 2.4

- **Covariate coefficients $\pm 0.75$:** The signs, set against the outcome coefficients listed under the outcome model below, create real confounding: $\sum_k \gamma_k \beta_k = 0.375$, so an analysis that ignores the confounders has a log-linear exposure slope biased by about 0.07 against a true slope of 0.10 (the confounders' combined effect on the outcome has correlation about 0.2 with $A_i$). With the sign pattern $(+, -, -, +)$ used by Josey et al. (2023) this sum is exactly zero, so the confounding there operates only through the exposure–covariate interaction
- **Note on skewness:** Ontario $PM_{2.5}$ is heavily right-skewed (skewness $\approx 9.2$, 99th percentile $\approx 26.4$, extreme values $> 200$ $\mu g/m^3$ during severe wildfire events). The Normal captures typical days but not extreme events.

**Choosing $\sigma_\phi^2$ via $R^2$ decomposition.** With the exposure model now defined, we can determine $\sigma_\phi^2$ by targeting a meaningful proportion of the total variance in $A_i$. The total variance decomposes into three sources: the covariate contribution $\mathrm{Var}(\mathbf{X}_i \text{ terms}) = 4 \times 0.75^2 = 2.25$, the spatial contribution $\mathrm{Var}(\phi_i)$, and the residual $\sigma_A^2 = 2.25$. Under the CAR prior the average marginal variance of $\phi_i$ is $\sigma_\phi^2 \, \bar{q}$, where $\bar{q} = \frac{1}{I} \mathrm{tr}\{(\mathbf{D}_W - \rho_A \mathbf{W})^{-1}\}$ is computed exactly from the grid. (The cruder approximation $\bar{q} \approx 1/\bar{d}$ with $\bar{d} \approx 3.6$ neighbours badly understates it at $\rho_A = 0.9$: a target of 0.30 gave a realised $R_\phi^2$ of about 0.50.) The proportion of variance explained by the spatial random effect is:

$$R_\phi^2 = \frac{\sigma_\phi^2 \bar{q}}{\mathrm{Var}(A_i)} = \frac{\sigma_\phi^2 \bar{q}}{9 + \sigma_\phi^2 \bar{q} + 9}.$$

Setting a target $R_\phi^2$ and solving gives $\sigma_\phi^2 = R_\phi^2 \times 18 / \{(1 - R_\phi^2)\, \bar{q}\}$. At the previously used value of $\sigma_\phi^2 = 0.3$, the implied $R_\phi^2 \approx 0.5\%$ — essentially no spatial contribution. We therefore parameterise through the target $R_\phi^2$, with base value 0.20, and compute the implied $\sigma_\phi^2$ accordingly; the realised share is about 0.20.

### 4.5 Measurement Error Model

Two parameterisations of the measurement error model are considered. Both generate $Z_i$ as a function of the true exposure $A_i$ and calibration covariate $D_i$, and both satisfy the non-differential assumption by construction since $Z_i$ does not appear in the outcome equation.

In both versions the measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ are assumed conditionally independent across areas given $(A_i, D_i)$. This is a simplifying assumption: Gryparis et al. (2009) show that prediction errors from spatial models are spatially correlated because nearby areas share the same monitoring station data when their surrogates are generated. Incorporating spatially correlated errors is listed as a potential extension.

**Version A: Classical error with Di-dependent mean (misspecification scenario).** Following Wu et al. (2019), $D_i$ enters through the mean of the surrogate, introducing systematic directional bias:

$$Z_i \mid A_i, D_i \sim \mathcal{N}(\beta_{Z1} A_i + \beta_{Z2} D_i,\ \sigma_Z^2). \tag{3}$$

This model is not fully identifiable when $A_i$ is latent: $\beta_{Z1}$ and $\sigma_Z^2$ cannot be separately recovered from the marginal distribution of $Z_i$ without validation data.

- $\beta_{Z1}$: attenuation. $\beta_{Z1} = 1$ centres the surrogate on the truth; $\beta_{Z1} < 1$ attenuates high exposures. Wu et al. (2019) use $\beta_1 = 0.8$ as their default.
- $\beta_{Z2}$: directional bias from $D_i$. When $\beta_{Z2} > 0$ the surrogate overestimates true exposure in high-$D_i$ areas (far from monitors), consistent with the MELONS finding of systematic overestimation in deprived and remote areas. When $\beta_{Z2} = 0$ this reduces to standard classical error.
- $\sigma_Z^2$: fixed homoscedastic noise. Heterogeneity across areas is captured entirely through the mean shift $\beta_{Z2} D_i$.

Wu et al. (2019) write this in the regression calibration direction as $E(X \mid W, D) = \gamma_0 + \gamma_1 W + \gamma_2^\top D$.

**Version B: Classical error with Di-dependent variance (primary model).** $D_i$ enters through the variance of the surrogate, relaxing the homoscedasticity assumption of Josey et al. (2023) Assumption 2:

$$Z_i \mid A_i, D_i \sim \mathcal{N}\big(\exp(\beta_1 D_i)\, A_i,\ \sigma_Z^2(D_i)\big), \qquad \sigma_Z^2(D_i) = \exp(\eta_0 + \eta_1 D_i). \tag{4}$$

The surrogate is not truncated; with $\eta_0 = -0.5$ a negative $Z_i$ is very rare (at most a couple of areas in occasional datasets) and is harmless because the Normal error model is exact.

- With $\eta_0 = -0.5$ the base value $\eta_1 = 0.5$ keeps the error SD moderate (median about 0.7, 99th percentile about 1.8, typical maximum about 2.7). At $\eta_1 = 1$ the right tail of $D_i$ would give a handful of areas a far larger error SD, and these areas would carry most of the total error variance.
- The calibration slope $\exp(\beta_1 D_i)$ is close to 1 on average ($\beta_1 = -0.1$ gives a mean slope of about 1.0) but falls below 1 in areas far from monitors (roughly 0.7 to 0.8 for $D_i > 1.5$), so the surrogate is attenuated and also noisier in high-$D_i$ areas. The log link keeps the slope positive despite the long right tail of $D_i$
- $\eta_1 = \beta_1 = 0$ recovers Josey's homoscedastic, unattenuated Assumption 2 exactly
- $\exp(\eta_0 + \eta_1 D_i)$ ensures positivity and implies a log-linear (proportional) relationship: a unit increase in $D_i$ multiplies the error variance by $\exp(\eta_1)$, natural for distance-based quantities (Gryparis et al., 2009)
- Unlike Version A, there is no directional shift in the mean; the departures from classical error are an attenuation and a noise magnitude that both depend on $D_i$
- This version is identifiable: the marginal variance of $Z_i$ given $(\mathbf{X}_i, D_i)$ is $\exp(2\beta_1 D_i)\,(\sigma_A^2 + \sigma_\phi^2 \bar{q}) + \exp(\eta_0 + \eta_1 D_i)$, allowing estimation of $(\eta_0, \eta_1)$ without observing $A_i$

### 4.6 Outcome Model

The area-level count outcome follows a Poisson model with a log-linear predictor and population offset, matching Josey et al. (2023):

$$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \text{Poisson}(N_i \exp(\eta_i)), \tag{5}$$

$$\eta_i = -3 + f(A_i) - 0.5X_{i1} - 0.25X_{i2} + 0.25X_{i3} + 0.5X_{i4}. \tag{6}$$

- **Population offset** $\log(N_i)$**:** $N_i$ scales the expected count so that $\exp(\eta_i)$ is the event rate per person. A larger area will have more events simply because it has more people; the offset removes this confound. In R this enters as an additive term $\log(N_i)$ in the linear predictor or via the `offset` argument.
- **Intercept** $-3$**:** $\exp(-3) \approx 5\%$ baseline event rate, representing baseline depression incidence in the CLSA population (Raina et al., 2019)
- **Confounder coefficients** $\pm(0.5, 0.25)$**:** Follow Josey et al. (2023)
- **ERF** $f(A_i)$**:** Exposure-response function specified in sensitivity scenario S4; three functional forms are evaluated
- **No spatial random effect:** Consistent with thesis proposal Section 4.3; spatial correlation in outcomes is induced indirectly through the spatially correlated $A_i$ surface

$Z_i$ does not appear in equation (5), satisfying the non-differential assumption by construction.

### 4.7 Causal Estimand

Following Josey et al. (2023), the target estimand is the marginal exposure-response function (ERF):

$$\theta(a) \equiv \mathbb{E}[Y_i(a)],$$

where $Y_i(a)$ is the potential outcome that would occur in area $i$ if, possibly contrary to what is observed, it had received exposure level $A_i = a$. The causal contrast between two exposure levels is $\Delta(a_1, a_0) = \theta(a_1) - \theta(a_0)$. Under the identifying assumptions in Section 3, the ERF is identified by the g-formula:

$$\theta(a) = \mathbb{E}[\mathbb{E}[Y_i \mid A_i = a, \mathbf{X}_i]]\,.$$

- The true $\theta(a)$ is computed by Monte Carlo using each replicate's own simulated covariates $\mathbf{X}_i$ (fresh per replicate, following Josey et al. 2023 `predict_example`), rather than a separate fixed external population. This lets the benchmark truth reflect the sampling variability of the covariate distribution itself alongside the estimators being evaluated
- **Evaluation grid:** the integers 5 to 15 (11 values), covering the bulk of the $A_i$ distribution (1st to 99th percentile about 4.6 to 15.4)
- **Primary causal contrast: extreme vs typical exposure:** $\Delta(Q_{99}, Q_{50})$ where $Q_{99}$ and $Q_{50}$ are the 99th and 50th percentiles of the simulated $A_i$ distribution. For the base model these correspond approximately to $Q_{50} \approx 10$ and $Q_{99} \approx 15.4$, representing the increase in expected depression case count on a severe wildfire day relative to a typical day. The percentiles are computed from the simulated $A_i$ values in each replicate so that the contrast automatically adapts to the exposure distribution under each scenario.
- **Secondary contrast: IQR:** $\Delta(Q_{75}, Q_{25})$ following standard practice in air pollution epidemiology

# 5 Simulation Parameter Grid

Table 1: Simulation parameters. Version A and Version B refer to the two error model parameterisations in Section 4. $R^2_\phi$ denotes the proportion of $\text{Var}(A_i)$ explained by the spatial random effect.

| Parameter | Notation | Base / Range | Justification |
|---|---|---|---|
| *Design* | | | |
| Number of areas | $I$ | 400 | $20 \times 20$ grid; Josey et al. (2023) use 400 and 800 clusters |
| Replicates | $R$ | 500 | |
| ERF evaluation points | | 11 (integers 5 to 15) | Covers the 1st to 99th percentile of $A_i$ |
| *Covariates* | | | |
| Area confounders | $X_{ik}$ | $X_{i2}$ standardised Gamma(2,1); others Normal | Mean 0, variance 1 |
| Population size | $\log(N_i)$ | $\mathcal{N}(8.5, 1.44)$ | Statistics Canada 2021 Census |
| Calibration covariate | $D_i$ | Gamma$(2, 1)$, standardised | Monitor distance proxy; Gryparis et al. (2009) |
| *True exposure* | | | |
| Mean | $\mu_{A,i}$ | $10 + 0.75(-X_{i1} + X_{i2} - X_{i3} + X_{i4})$ | Mean 10 $\mu$g/m$^3$; keeps $A_i$ and $Z_i$ non-negative without truncation |
| SD | $\sigma_A$ | 1.5 | Residual SD; total SD of $A_i$ about 2.4 |
| Spatial $R^2$ | $R^2_\phi$ | 0.20 | Proportion of $\text{Var}(A_i)$ explained by spatial effect (exact CAR calibration) |
| Spatial correlation | $\rho_A$ | 0.9 | Base; $\rho_A = 0$ used only as a no-spatial check |
| *Error model Version A* | | | |
| Attenuation | $\beta_{Z1}$ | 0.8 | Wu et al. (2019) default |
| Directional bias | $\beta_{Z2}$ | $\{0, 2\}$ | $0$ = no bias; $> 0$ = overestimation |
| Noise variance | $\sigma^2_Z$ | 1 | Fixed moderate noise |
| *Error model Version B* | | | |
| Baseline variance | $\eta_0$ | $-0.5$ | $\exp(-0.5) \approx 0.61$ at $D_i = 0$; reliability $\text{Var}(A)/\text{Var}(Z) \approx 0.77$ |
| Di-slope | $\eta_1$ | 0.5 | Moderate heteroscedasticity; $\eta_1 = 0$ recovers Josey |
| Attenuation slope | $\beta_1$ | $-0.1$ | Calibration slope $\exp(\beta_1 D_i)$: about 1 on average, roughly 0.7 to 0.8 far from monitors |
| *Outcome* | | | |
| Intercept | | $-3$ | $\approx 5\%$ baseline event rate (CLSA) |
| ERF | $f(A_i)$ | Linear/Quadratic/Nonlinear | |
| Confounder coefs | $\boldsymbol{\gamma}$ | $(-0.5, -0.25, 0.25, 0.5)$ | |
| *Sensitivity scenarios* | | | |
| $D_i^*$ distribution | | Exp(1); empirical | Monotone decreasing; real Ontario data |
| $N_i$ distribution | | Uniform$(10, 1000)$ | Josey comparability |
| Exposure distribution | | Log-normal | Extreme wildfire events |
| Number of areas | $I$ | $\{196, 900\}$ | Scaling; all perfect squares |

# 6 Sensitivity Scenarios

$D_i$ **distribution. Gamma(2, 0.5)**: mean 4, more heterogeneity between areas. **Exponential(1)**: monotone decreasing, consistent with Southern Ontario station concentration. **Empirical**: actual FSA-to-NAPS distances from Statistics Canada shapefiles.

**Error model alternatives. Version A vs Version B**: Data generated under Version A (directional bias, $\beta_{Z2} \in \{0, 1, 2\}$) but analysed with a Version B error model. Quantifies performance degradation when the true error has directional bias that the proposed method does not model. $\beta_{Z2} = 0$ provides an internal consistency check.

**Spatial correlation. Scenario B ($\rho_A = 0$)**: spatially independent exposures. Directly comparable to Josey et al. (2023) which sets $\phi_i = 0$ for all areas. **Scenario A ($\rho_A = 0.9$)**: high spatial smoothing, reflecting a sustained regional wildfire event where nearby FSAs share similar smoke concentrations.

**Exposure-response functional forms.** Three forms spanning simple to complex, crossed with spatial scenarios to give six main simulation cells.

**Linear**: $f(A_i) = 0.10(A_i - 10)$. Monotone increasing with no curvature.

**Quadratic**: $f(A_i) = 0.30(A_i - 10) - 0.05(A_i - 10)^2$. Concave shape that rises then declines within the evaluation window (vertex at $A_i = 13$).

**Nonlinear (up–down–up)**: $f(A_i, X_{i1}) = 0.12(A_i - 10) + 0.55\big[\sin\big(2\pi(A_i - 6.25)/9\big) - 0.5\big] - 0.25\tanh\big((A_i - 10)/4\big)X_{i1}$, centred so that $f(10, \cdot) = 0$ like the other forms. A linear trend plus one sine wave: the curve rises to a local maximum near $A_i = 9$, falls to a local minimum near $A_i = 12.5$, and then rises again through the end of the window. It is not captured by a linear or quadratic form, and it grows only linearly beyond the window (exposure rarely exceeds 17), unlike a cubic. The bounded interaction with $X_{i1}$ (at most 0.25 per SD) makes the exposure effect depend on $X_{i1}$, in the spirit of Josey et al. (2023) but without their unbounded $(A_i - 10)X_{i1}$ term; the marginal ERF is then a genuine average of different conditional curves.

**Log-normal exposure.** $\log(A_i) \sim \mathcal{N}(1.68, 0.54)$ matching Ontario moments: median 5.4, mean 7, SD 5.9. Captures extreme wildfire events (99th percentile $\approx 26$ $\mu$g/m$^3$) that the Normal cannot represent.

# 7 Performance Metrics

For each scenario and method (naive, oracle, proposed, regression calibration):

- **Bias**: $\hat{\theta}(a) - \theta(a)$ averaged over evaluation points and replicates
- **RMSE**: $\sqrt{R^{-1}\sum_r(\hat{\theta}_r(a) - \theta(a))^2}$
- **Coverage**: fraction of 95% credible intervals containing $\theta(a)$

- **Interval width**: mean width of 95% credible intervals
- **Causal contrast bias**: $\hat{\Delta}(Q_{99}, Q_{50}) - \Delta(Q_{99}, Q_{50})$ and $\hat{\Delta}(Q_{75}, Q_{25}) - \Delta(Q_{75}, Q_{25})$, where quantiles are computed from the simulated $A_i$ distribution in each replicate

# 8 Potential Extensions

- **Spatially correlated measurement errors:** The current model assumes conditional independence of measurement errors $\varepsilon_i = Z_i - \mathbb{E}[Z_i \mid A_i, D_i]$ across areas which is a simplifying assumption that Gryparis et al. (2009) show is violated in practice: prediction errors from spatial models are correlated across nearby areas because those predictions are generated using the same monitoring station data. A natural extension is to give the error term its own spatial covariance structure, either by making $D_i$ spatially structured via a second CAR prior (so that nearby areas have correlated error variances), or by adding an explicit spatially correlated residual $\psi_i \sim \text{CAR}(\rho_\varepsilon, \sigma_\varepsilon^2)$ to the error model.
- **Individual-level binary outcome:** Replacing the area-level Poisson count with individual-level Bernoulli outcomes $Y_{ij} \sim \text{Bernoulli}(p_{ij})$ and individual covariates $L_{ij}$, matching the CLSA data structure. The Bayesian bootstrap would draw Dirichlet weights over all $N = \sum_i n_i$ individuals rather than over areas.
- **Spatial random effect on outcome ($\nu_i$)**: A second CAR prior capturing residual spatial clustering in health outcomes (Englert et al., 2025).
- **Negative binomial outcome:** The Poisson model assumes $\text{Var}(Y_i) = \mathbb{E}[Y_i]$, which is often violated in practice when area-level counts are overdispersed due to unmeasured heterogeneity across FSAs. A negative binomial outcome relaxes this by introducing a dispersion parameter $\xi > 0$:

  $$Y_i \mid A_i, \mathbf{X}_i, N_i \sim \text{NegBin}(\mu_i, \xi), \qquad \mu_i = N_i \exp(\eta_i),$$

  As $\xi \to \infty$ this recovers the Poisson; small $\xi$ gives strong overdispersion. The dispersion parameter $\xi = 1$ is a natural starting point and is the value used by Englert et al. (2025) in their simulation. This extension tests whether the proposed method remains well-calibrated when the outcome model is more variable than the base Poisson.
- **Zero-inflated outcome**: For rare outcomes or small populations where structural zeros arise.
- **Log-normal exposure model:** The base model specifies $A_i$ on the original scale as a Normal with a floor at zero. An alternative motivated by the right-skewed nature of $PM_{2.5}$ concentrations and by the suggestion that $\log(A_i)$ should be regressed on covariates is to model the log-exposure directly:

  $$\log(A_i) \mid \mathbf{X}_i, \phi_i \sim \mathcal{N}\big(\gamma_0 + \gamma_1 X_{i1} + \gamma_2 X_{i2} + \gamma_3 X_{i3} + \gamma_4 X_{i4} + \phi_i,\ \sigma^2_{\log A}\big)\,,$$

  so that $A_i = \exp(\log A_i) > 0$ always. The parameters are matched to Ontario $PM_{2.5}$ empirical moments: setting $\gamma_0 = 1.68$ and $\sigma_{\log A} = 0.73$ gives $\mathbb{E}[A_i] \approx 7\ \mu\text{g/m}^3$ and $\text{SD}(A_i) \approx 5.9\ \mu\text{g/m}^3$, matching the provincial monitoring data. The error model in Version B would then operate on the log scale as $\log(Z_i) \mid \log(A_i), D_i \sim \mathcal{N}(\log(A_i), \exp(\eta_0 + \eta_1 D_i))$, giving multiplicative error on the original scale.